



DGX Spark User Guide

NVIDIA Corporation

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This guide is also available for download as a [PDF](#).

Chapter 1. Overview

The DGX Spark is NVIDIA's compact AI computer designed for developers, data scientists, and AI researchers who need powerful computing capabilities for AI development and deployment.

Chapter 2. Getting Started

2.1. DGX Spark Release Notes

This section provides release notes for the DGX Spark, including information about new features, known issues, and software version updates.

2.1.1. Current Software Versions

The following table shows the current version information for the DGX Spark software stack.

Component	Version
NVIDIA DGX OS	7.4.0
NVIDIA GPU Driver	580.126.09
NVIDIA CUDA Toolkit	13.0.2
Canonical Kernel	6.17

2.1.1.1 January 2026 Release

2.1.1.1.1 What's New

- ▶ **Power Management** - Added hot plug support for the ConnectX-7 network adapter, saving up to 18W of power when the adapter is not in use.
- ▶ **Bluetooth Audio Support** - Support for Bluetooth audio devices, including headphones and headsets, is now available.
- ▶ **Additional Security Controls** - WiFi and Bluetooth can now be disabled in the Unified Extensible Firmware Interface (UEFI) for environments requiring additional security.

2.1.1.1.2 Fixed Issues

- ▶ **Improved Monitor Support** - Better compatibility with TVs and monitors, including multi-monitor configurations and the use of non-native resolutions and refresh rates.
- ▶ **Improved Peripheral Setup** - Smoother and more complete Out of Box Experience (OOBE) navigation.

2.1.1.2 November 2025 Release

2.1.1.2.1 What's New

- ▶ **New DGX OS kernel** - The operating system now incorporates the Ubuntu 6.14 Hardware Enablement (HWE) kernel stack. The new kernel brings ongoing performance gains, enhanced stability, broader hardware compatibility, and the latest security updates for a more secure operating environment.
- ▶ **JupyterLab updated to latest CUDA and PyTorch** - Updated JupyterLab to CUDA 13.0.2 and the latest PyTorch version, enabling users to work with updated frameworks immediately without extra downloads.

2.1.1.2.2 Fixed Issues

- ▶ **Improved memory reporting in DGX Dashboard** - Addressed the issue with memory reporting differences with unified memory architecture; readout now consistent with CUDA guidance for unified memory systems. See https://nvidia.custhelp.com/app/answers/detail/a_id/5728.
- ▶ **Image Generation in JupyterLab** - Resolved the inability to generate images in Stable Diffusion XL Playbook example. Users can now run end-to-end example workflow inside JupyterLab.
- ▶ **Improved Peripheral Interoperability** - Better compatibility with USB-C devices, monitors, Bluetooth peripherals, and Wi-Fi access points.
- ▶ **Improved recovery image reliability** - The recovery image now installs correctly on macOS and also when multiple external USB-C drives are connected.
- ▶ **Keyboard accessibility improvements** - Smoother and more complete OOB navigation, including the ability to complete entire system setup process using only keyboard.

2.1.2. Known Issues

For an updated list of known issues, see [Known Issues](#).

2.2. Known Issues

This topic provides a summary of known issues with DGX Spark systems.

2.2.1. Use the supplied power adapter for optimal performance

For optimal performance, use the supplied power adapter with the DGX Spark system. Using a different adapter may reduce performance, prevent boot, or cause unexpected shutdowns.

2.2.2. `nvidia-smi` reports “Memory-Usage: Not Supported”

On iGPU platforms, `nvidia-smi` will display “Memory-Usage: Not Supported” even though per-process GPU memory is listed. This is expected because iGPUs do not have dedicated framebuffer

memory.

2.2.3. CUDA Support on DGX Spark

The CUDA version on your DGX Spark device has been verified to work with your system hardware at the time of software update release. The latest features and performance improvements are available through NVIDIA NGC containers (for example, PyTorch, vLLM, and TensorRT-LLM).

2.2.4. Guidance for reporting memory resources with unified memory architecture

NVIDIA is actively working with third-party ecosystem partners to bring their software to DGX Spark. For example, we have provided direction on implementing memory management on systems based on a unified memory architecture (UMA) to help ensure accurate reporting of available resources.

DGX Spark systems use a unified memory architecture (UMA), where the GPU shares system memory (DRAM) with the CPU and other compute engines. This design reduces latency and allows larger amounts of memory to be used for GPU workloads. On UMA systems, the CPU can dynamically manage DRAM contents, including freeing up memory by swapping pages between DRAM and the system's SWAP area. However, the `cudaMemGetInfo` API does not account for memory that could potentially be reclaimed from SWAP. As a result, the memory size reported by `cudaMemGetInfo` may be smaller than the actual allocatable memory, since the CPU may be able to release additional DRAM pages by moving them to SWAP.

To more accurately estimate the amount of allocatable device memory on DGX Spark platforms, CUDA application developers should consider the possibility of DRAM reclamation via SWAP and not rely solely on the values returned by `cudaMemGetInfo`. The following provides an example implementation using C standard libraries:

```
#include <stdio.h>

int getAvailableMemory(long *availableMemoryKb, long *freeSwapKb) {
    FILE *meminfoFile = NULL;
    char lineBuffer[256];
    long hugeTlbTotalPages = -1;
    long hugeTlbFreePages = -1;
    long hugeTlbPageSize = -1;

    if (availableMemoryKb == NULL || freeSwapKb == NULL) {
        return 1;
    }

    meminfoFile = fopen("/proc/meminfo", "r");
    if (meminfoFile == NULL) {
        return 1;
    }

    *availableMemoryKb = -1;
    *freeSwapKb = -1;

    while (fgets(lineBuffer, sizeof(lineBuffer), meminfoFile)) {
```

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```

    long value;

    if (sscanf(lineBuffer, "MemAvailable: %ld kB", &value) == 1) {
        *availableMemoryKb = value;
    } else if (sscanf(lineBuffer, "SwapFree: %ld kB", &value) == 1) {
        *freeSwapKb = value;
    } else if (sscanf(lineBuffer, "HugePages_Total: %ld", &value) == 1) {
        hugeTlbTotalPages = value;
    } else if (sscanf(lineBuffer, "HugePages_Free: %ld", &value) == 1) {
        hugeTlbFreePages = value;
    } else if (sscanf(lineBuffer, "Hugepagesize: %ld kB", &value) == 1) {
        hugeTlbPageSize = value;
    }

    if (*availableMemoryKb != -1 &&
        *freeSwapKb != -1 &&
        hugeTlbTotalPages != -1 &&
        hugeTlbFreePages != -1 &&
        hugeTlbPageSize != -1) {
        break;
    }
}

fclose(meminfoFile);

if (hugeTlbTotalPages != 0 && hugeTlbTotalPages != -1) {
    *availableMemoryKb = hugeTlbFreePages * hugeTlbPageSize;

    // Hugetlbfs pages are not swappable.
    *freeSwapKb = 0;
}

return 0;
}

```

As a workaround for debugging purposes, you can flush the buffer cache manually with the following command:

```
sudo sh -c 'sync; echo 3 > /proc/sys/vm/drop_caches'
```

After flushing the cache, restart your application.

2.2.5. Attached HDMI display goes into deep sleep mode after an extended period of inactivity and does not wake

In some cases, after a long period of inactivity, the attached HDMI display might go into a deep sleep mode and not wake up when you press a key or move the mouse. If this happens, press a physical button on the monitor to wake it up.

2.3. Hardware Overview

Powered by the NVIDIA Grace Blackwell architecture, DGX Spark enables developers, researchers, and data scientists to prototype, deploy, and fine-tune large AI models on their desktop. This section provides information about the hardware components and specifications.

2.3.1. System Overview

The DGX Spark features:

- ▶ NVIDIA Grace Blackwell architecture with integrated GPU and CPU
- ▶ 20-core Arm processor with high-performance cores
- ▶ 128 GB unified system memory
- ▶ Compact desktop form factor
- ▶ Advanced connectivity including Wi-Fi 7, 10 GbE, and ConnectX-7
- ▶ Support for AI models up to 200 billion parameters (or 405B for dual-Spark configuration)

2.3.1.1 Component Descriptions

The DGX Spark includes the following components:

Table 1: Component Specifications

Component	Specification
GPU	NVIDIA Blackwell Architecture with 5th Generation Tensor Cores, 4th Generation RT Cores
CPU	20-core Arm processor (10 Cortex-X925 + 10 Cortex-A725)
Memory	128 GB LPDDR5x unified system memory, 256-bit interface, 4266 MHz, 273 GB/s bandwidth
Storage	1 TB or 4 TB NVMe M.2 with self-encryption
Network	1x RJ-45 (10 GbE), ConnectX-7 Smart NIC, Wi-Fi 7, Bluetooth 5.4
Connectivity	4x USB Type-C, 1x HDMI 2.1a, HDMI multichannel audio
Video Processing	1x NVENC, 1x NVDEC

2.3.2. Physical Specifications

2.3.2.1 Form Factor

- ▶ **Chassis Type:** Small form factor (SFF)
- ▶ **Dimensions:** 150 mm (L) x 150 mm (W) x 50.5 mm (H)
- ▶ **Weight:** 1.2 kg (2.6 lbs)

2.3.2.2 Environmental Requirements

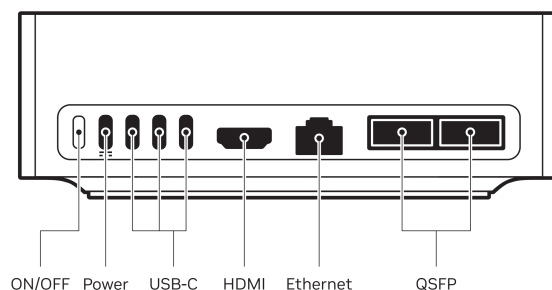
Table 2: Environmental Specifications

Specification	Value
Ideal Operating Temperature	5°C to 30°C (41°F to 86°F)
Operating Humidity	10% to 90% (non-condensing)
Operating Altitude	Up to 3,000 meters (9,843 feet)

2.3.3. Connectivity and I/O

2.3.3.1 Rear Panel

- ▶ Power button
- ▶ 4x USB Type-C (one for power delivery)
- ▶ 1x HDMI 2.1a display connector
- ▶ 1x RJ-45 Ethernet connector (10 GbE)
- ▶ 2x QSFP Network connectors (ConnectX-7)



2.3.4. Performance Specifications

2.3.4.1 Compute Performance

- ▶ **AI Compute:** Up to 1,000 TOPS (trillion operations per second) inference and up to 1 PFLOP (petaFLOP) at FP4 precision with sparsity
- ▶ **CUDA Cores:** 6,144
- ▶ **Copy Engines:** 2 (enables simultaneous data transfers to and from GPU memory, improving throughput for AI workloads)
- ▶ **CPU Performance:** 20 cores (10 Cortex-X925 + 10 Cortex-A725)
- ▶ **Memory Bandwidth:** 273 GB/s
- ▶ **Memory Channels:** 16 channels (256 bit) LPDDR5X 8533

2.3.4.2 AI/ML Capabilities

- ▶ **Model Support:** AI models up to 200 billion parameters
- ▶ **Tensor Performance:** 5th Generation Tensor Cores with FP4 support
- ▶ **Framework Support:** PyTorch, TRT-LLM, and other AI frameworks
- ▶ **Use Cases:** Inference, deployment, and fine-tuning of large language models

2.3.5. Power and Thermal Management

2.3.5.1 Power Requirements

- ▶ **Power Supply:** 240W external power supply (included)
 - ▶ GB10 SOC Thermal Design Power (TDP) is 140W
 - ▶ 100W is available for other system components (ConnectX-7, Wi-Fi, SSD, USB-C, etc.)
- ▶ **Usage Requirement:** Use of the provided 240W power supply is required for optimal performance. Using a different or lower-rated power supply may result in reduced system performance, failure to boot, or unexpected shutdowns.
- ▶ **Input Voltage:** Standard AC power input

2.3.5.2 Thermal Management

- ▶ **Cooling Solution:** Integrated thermal management system
- ▶ **Form Factor:** Compact design optimized for desktop placement
- ▶ **Ideal Operating Temperature:** 5°C to 30°C (41°F to 86°F)

2.4. Initial Setup - First Boot

This guide walks you through setting up your DGX Spark for the first time. You'll choose how to access your system during the initial setup, and run the first-time setup utility to configure everything. The access method you choose is only for completing the initial setup - after setup is complete, you can access your DGX Spark any way you like: locally with a monitor and keyboard, over the local network from another computer, or a mix of both.

2.4.1. What You'll Do

This setup process includes:

- ▶ Choosing how to access the system during initial setup (with a display, or as a network appliance)
- ▶ Preparing your system and connections
- ▶ Running the first-time setup utility to configure your system

2.4.2. Choose how to access your system during initial setup

To complete the initial setup, you'll need to access your DGX Spark. You can do this in one of two ways:

With a Display (Local Setup)

- ▶ Connect keyboard and mouse via USB or Bluetooth
- ▶ Connect a display to work directly on the system
- ▶ Follow the setup wizard on screen

Over the Network (as a Network Appliance)

- ▶ Access the system over your local network from another computer
- ▶ Use another computer to complete setup via web browser
- ▶ No Spark display or keyboard required for the setup process

Note

This choice is only about how you'll complete the initial setup process. After setup is finished, you can access your DGX Spark however you prefer. You're not locked into your original choice.

2.4.3. Get Ready

Important

The DGX Spark device starts up immediately when power is applied. Please attach all peripherals (display, keyboard, mouse, network, etc.) before connecting the power supply.

Before starting, ensure you have:

- ▶ A fast, reliable internet connection (Wi-Fi or Ethernet) to download required updates during the initial setup process. Connections using captive portals (such as hotel or airport Wi-Fi) or those prone to disconnections (like phone hotspots) are not recommended. If you do not have access to a stable connection, consider downloading the system recovery media and using [System Recovery](#) to install the latest software for your DGX Spark.
- ▶ For Local Setup: A display, keyboard, and mouse connected (or available using Bluetooth).
- ▶ For Network Setup: A computer on the same network to access the setup interface.
- ▶ Power connected to the system (the system will start automatically when power is applied).

Note

Display Troubleshooting: Some displays may have trouble with Spark out of the box. If you are connecting over USB-C/DisplayPort and there is no display, try using HDMI instead.

Note

If you plan to use a wired network connection, plug in the network cable before starting the installation. This helps avoid connection issues later in the process.

2.4.4. Run the First-Time Setup

The first-time setup utility will guide you through:

- ▶ Powering on and initializing the system
- ▶ Selecting your preferred setup mode
- ▶ Downloading and installing critical updates
- ▶ Completing your initial configuration

Warning

Critical: Do not shut down or reboot the system during the update process. The installation cannot be interrupted once the download begins, and powering down during updates can cause system damage.

2.4.4.1 Getting Started

The way you start the installation depends on your chosen access method:

With a Display (Local Setup)

1. Power on the system
2. The first-time setup utility will start automatically on the connected display
3. Use your wired keyboard and mouse (already connected) to navigate
4. If a keyboard or mouse is not detected, you will be prompted to put your Bluetooth devices in pairing mode:

USB devices can be plugged in at any time and should start working, even if detected improperly. Bluetooth devices can be put into pairing mode and will generally still pair while on the “Get Started” screen (exception - keyboards that require a passcode to type in won’t work on this screen). Once you click on “Get Started,” Bluetooth pairing stops, so you will have to power cycle to try again.
5. Follow the on-screen prompts to complete the setup process

Over the Network (as a Network Appliance)

1. Power on the system. This creates a Wi-Fi hotspot that you will use to connect to the system and continue the setup process. The SSID and password for the Wi-Fi hotspot are printed on a sticker attached to the Quick Start Guide included with your DGX Spark’s packaging
2. From another computer, connect to the Spark’s Wi-Fi hotspot using the SSID and password provided on the Quick Start Guide. A captive portal page will open in the default web browser on your computer. If it does not open automatically, use your browser to navigate to the Spark’s system setup page listed on the Quick Start Guide.

3. Follow the on-screen prompts to continue the setup process.

When the DGX Spark joins your home network, its Wi-Fi hotspot will turn off and your computer will reconnect to the device through your Wi-Fi network to resume the setup process. If you are not able to connect to the DGX Spark after it joins your home network, you must connect a display/keyboard/mouse to continue.

Note

It may take as long as 10 minutes for the DGX Spark to install all of its updates and join your home network.

2.4.4.2 What to Expect During Setup

The first-time setup utility will guide you through several configuration steps. Simply follow the on-screen prompts to complete each step.

Setup Process Steps:

1. **Language and Time Zone Selection**

Choose your preferred language and time zone settings for the system. Note that the input fields will filter as you type.

2. **Keyboard Layout Selection** (Local Setup only)

Select your keyboard layout (e.g. US keyboard vs. Russian keyboard). This screen only appears when using a display during setup.

3. **Terms and Conditions**

Review and accept the terms and conditions to continue with the installation.

4. **User Account Creation**

Create your username and password for system access.

5. **Information Sharing Settings** (Optional)

Configure analytics and crash reporting preferences. You can skip this step if desired.

6. **Wi-Fi Network Selection**

Select your Wi-Fi network. This step is automatically skipped if an Ethernet cable that is providing internet access is connected.

7. **Wi-Fi Password**

Enter the password for your selected Wi-Fi network.

8. **Joining Wi-Fi Network**

The system connects to your Wi-Fi network and tears down the access point. Your computer will automatically reconnect to your default network.

Note**Network Connection Issues:**

- If your computer automatically reconnects to the same network as the Spark, the installation should continue seamlessly

- ▶ If not, you'll need to connect your computer to the same network as the Spark while the setup app is waiting for the network setup process to complete
- ▶ If the setup fails, you must connect a display/keyboard/mouse to continue
- ▶ The modal instructs you to try to reconnect to the Spark's hotspot and try again. This will work if the Spark actually failed to join the network (e.g. wrong password) versus your laptop can't communicate with the Spark
- ▶ If you DO NOT see the hotspot available when this error modal appears, that means the Spark did join the network but your laptop can't communicate with it. This could be because:
 - ▶ Device isolation
 - ▶ You failed to join the same network as your Spark
 - ▶ mDNS does not work on your network due to its configuration (e.g. a complex corporate network)

9. Software Download and Installation

After connecting to the network, the system will automatically download and install the full software image. This process can take some time, and the system may reboot more than once before it's complete. If you are setting up over the network from another computer, you will not be able to access the device during this time. The process may continue for up to 10 minutes after the interface shows that the device is rebooting to finish the setup

Warning

Do not shut down or reboot the system during this process. The installation cannot be interrupted once the download begins.

10. Installation Complete

The device will reboot automatically when installation is complete, and you can then use it normally.

2.4.5. Next Steps

Congratulations! Your DGX Spark is now ready to use. Here are the recommended ways to access and start working with your system:

Access Your System

- ▶ **Locally:** Use your DGX Spark like any desktop computer with a monitor, keyboard, and mouse
- ▶ **Over the Network:** Connect to your DGX Spark from another computer on the same network using the NVIDIA Sync tool (see [NVIDIA Sync](#)), SSH, or remote desktop
- ▶ **Mix Both:** Use whatever access method fits your current workflow - you can switch freely between local and network access
- ▶ **Dashboard Access:** Use the built-in DGX Dashboard for system monitoring, updates, and JupyterLab access. See [DGX Dashboard](#) for more information

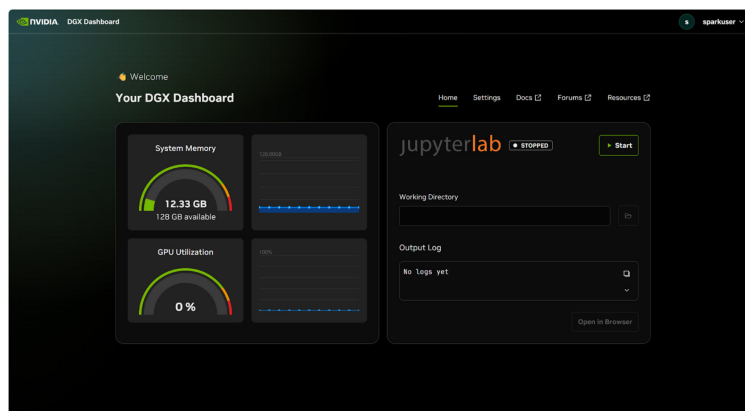


Fig. 1: The DGX Dashboard provides an intuitive interface for system monitoring and development

Additional Resources

- ▶ Visit the [NVIDIA Spark Developer Portal](https://build.nvidia.com/spark) at <https://build.nvidia.com/spark> for the latest guides, tutorials, and updates
- ▶ Refer to [DGX Spark Release Notes](#) for the latest software updates and features
- ▶ See [Known Issues](#) for troubleshooting common problems

Your DGX Spark is now ready to power your AI development and deployment workflows!

2.5. System Configuration and Operation

Configuring and operating your DGX Spark effectively is key to delivering consistent results across AI/ML workflows. This section provides an overview of the platform, recommended UEFI settings, clustering procedures, and foundational security guidance to help you deploy, manage, and scale with confidence.

2.5.1. System Overview

Powered by the NVIDIA Grace Blackwell architecture, DGX Spark enables developers, researchers, and data scientists to prototype, deploy, and fine-tune large AI models on their desktop.

2.5.1.1 Flexible Access and Usage

The DGX Spark is designed for maximum flexibility in how you access and use it. You can seamlessly switch between different access methods based on your needs:

- ▶ **Local Access:** Connect a keyboard, mouse, and monitor to work directly on the system
- ▶ **Network Access:** Access your system from another computer on the same network using SSH, NVIDIA Sync, or remote desktop tools
- ▶ **Hybrid Usage:** Mix and match access methods - work locally one day and over the network the next, or even simultaneously

All access methods are fully supported and equally capable. Your DGX Spark adapts to your workflow, whether you're working at your desk with a monitor or accessing it remotely as a network appliance on the same network.

2.5.1.2 Key Capabilities

Your DGX Spark enables you to:

- ▶ **Run Inference:** Deploy models for real-time AI applications
- ▶ **Develop AI Models:** Train and fine-tune models with up to 200 billion parameters
- ▶ **Process Data:** Handle large datasets with high-performance computing
- ▶ **Experiment Freely:** Test new ideas without cloud computing costs
- ▶ **Scale Workloads:** Connect multiple systems for larger projects

2.5.1.3 System Architecture

The DGX Spark is built on NVIDIA's Grace Blackwell architecture, providing:

- ▶ **Unified Memory:** 128 GB of high-bandwidth memory for large models
- ▶ **High-Performance Computing:** 20-core ARM64-based processor with integrated GPU
- ▶ **Advanced Connectivity:** Wi-Fi 7, 10 GbE, CX7 NIC, and multiple I/O options
- ▶ **Compact Form Factor:** 150mm x 150mm x 50.5mm desktop design

For detailed hardware specifications, see [Hardware Overview](#).

2.5.1.4 Software

Your system comes pre-configured with:

- ▶ **NVIDIA DGX OS:** Optimized operating system for AI workloads
- ▶ **Development Tools:** CUDA, cuDNN, and NVIDIA's development ecosystem
- ▶ **Container Support:** Docker and NVIDIA Container Runtime for easy deployment
- ▶ **NGC Integration:** Access to NVIDIA's container registry

For detailed software information, see [Software](#).

2.5.1.5 Getting Started

To begin using your DGX Spark:

1. **Initial Setup:** Follow the [Initial Setup - First Boot](#) to configure your system
2. **Explore Examples:** Try sample workloads to understand capabilities
3. **Configure Development Environment:** Set up your preferred tools and frameworks
4. **Start Building:** Begin your AI development projects

Note

For the most up-to-date tutorials and examples, visit <https://build.nvidia.com/spark>. This site is regularly updated with new content and serves as the primary resource for practical guides and use cases.

2.5.2. UEFI Settings

This topic provides guidance on accessing and configuring the UEFI settings for the DGX Spark system. While there are no Spark-specific features that require UEFI configuration, you may need to access the UEFI for general system configuration or troubleshooting purposes.

2.5.2.1 Access UEFI

Important

To access the UEFI setup menu, you must be using a keyboard that is physically connected to the DGX Spark device.

If the keyboard is a Mac keyboard, it might not have a key that UEFI recognizes as Del; use Esc only.

To access the UEFI setup menu:

1. Power on or restart the system.
2. Immediately press Esc or Del and hold it until the UEFI setup menu appears.

Note

The timing for pressing Esc or Del is critical. Press the key as soon as the system starts booting, before the OS begins loading.

2.5.2.2 UEFI Documentation

For detailed information about UEFI settings and configuration options, refer to the [DGX Spark UEFI Manual](#).

2.5.2.3 Enable or Disable WiFi and Bluetooth

You can enable or disable both Wireless LAN and Bluetooth together in UEFI, which completely disables all wireless access for security enhancement. To disable only one (WiFi or Bluetooth) individually, use the OS settings instead. You might disable wireless to meet lab or security policy, or enable it for initial setup when Ethernet is not available.

1. Go to **Advanced** ▸ **Advanced Menu** ▸ **IO Port Access**.
2. Set **Wireless LAN & Bluetooth** to the desired state (enabled or disabled).

2.5.2.4 Boot from a USB Device

To boot from a USB device, use one of the following methods. You might do this to run system recovery, reinstall the OS, or boot from live or diagnostic media.

Set USB as the first boot device (persistent):

1. Go to **Boot** ▸ **Boot Option Priorities**.
2. Use Enter on each entry to set the boot order. Place the USB device at the top.

Boot from USB once (one-time):

1. Go to **Save & Exit**.

2. Under **Boot Override**, select the USB device and press Enter. The system boots from that device for the current boot only.

The USB device must appear in the boot option list. If it does not, ensure the device is connected before you enter UEFI and that it is bootable.

2.5.2.5 Enable or Disable Secure Boot

To enable or disable Secure Boot, perform the following steps. You might disable it to use PXE boot or run custom boot loaders, or enable it to meet security or compliance requirements.

1. Go to **Security** ▢ **Secure Boot**.
2. Set **Secure Boot** to **Enabled** or **Disabled** as needed. The default is **Enabled**.
3. Save changes and exit from the **Save & Exit** menu. A platform reset may be required for the change to take effect.

Secure Boot is active when it is enabled, the Platform Key (PK) is enrolled, and the system is in User mode.

2.5.2.6 Configure PXE Boot

To configure UEFI for PXE boot, perform the following steps. You might do this to deploy or reimagine the DGX Spark over the network, or to boot the recovery image from a PXE server.

Note

Before enabling PXE boot, you must either disable Secure Boot or enroll the PXE bootloader (grubnetaa64.efi.signed) in UEFI. For details, see [PXE Boot Setup](#).

1. Go to **Advanced** ▢ **Network Stack Configuration**.
2. Set **Network Stack** to **Enabled**.
3. Set **Ipv4 PXE Support** or **Ipv6 PXE Support** (or both) to **Enabled** as needed.
4. Optionally set **PXE boot wait time** (seconds to press Esc to abort PXE) and **Media detect count**.
5. In **Advanced**, open the network configuration screen for the NIC you want to use for PXE (listed by MAC address). Enable and configure IPv4 or IPv6 for that device as required.
6. Save changes and exit from the **Save & Exit** menu.

For PXE server setup, bootloader configuration, and DHCP setup, see [PXE Boot Setup](#).

Important

Always save your UEFI configuration changes from the **Save & Exit** menu so the system restarts with your new settings applied.

2.5.2.7 Troubleshooting

If you experience issues after making UEFI changes:

- Revert to UEFI defaults and restart
- Apply changes incrementally to identify problematic settings

- ▶ Update to the latest UEFI version if available
- ▶ Consult the [DGX Spark UEFI Manual](#) for specific setting descriptions
- ▶ For additional troubleshooting guidance and support options, see [spark-maintenance-troubleshooting](#).

2.5.3. Spark Stacking

2.5.3.1 Connecting DGX Spark Systems into a Virtual Cluster

2.5.3.1.1 Overview

This guide explains how to connect two DGX Spark systems into a virtual compute cluster using simplified networking configuration and a QSFP/CX7 cable for high-performance interconnect.

The goal is to enable distributed workloads across Grace Blackwell GPUs using MPI (for inter-process CPU communication) and NCCL v2.28.3 (for GPU-accelerated collective operations).

Additional Information can be found in the [Connect Two Sparks](#) playbook.

2.5.3.1.2 System Requirements

Before you begin, ensure the following:

- ▶ Both DGX Spark systems have Grace Blackwell GPUs, are connected to each other using a QSFP/CX7 cable, and are running Ubuntu 24.04 (or later) with NVIDIA drivers installed

Note

The DGX Spark CX-7 ports support ethernet configuration only.

Approved cables for the CX-7 ports are:

- ▶ Amphenol: NJAAKK-N911 (QSFP to QSFP112, 32AWG, 400mm, LSZH), *NJAAKK0006 is the 0.5m version of this cable*
- ▶ Luxshare: LMTQF022-SD-R (QSFP112 400G DAC Cable, 400mm, 30AWG)

- ▶ The systems have internet access for initial software setup
- ▶ You have sudo/root access on both systems

2.5.3.1.3 Setup Networking Between Spark Systems

2.5.3.1.3.1 Option 1: Automatic IP assignment (recommended)

Follow these steps on **both DGX Spark nodes** to configure network interfaces using netplan. The following commands should be run in a terminal session (either local or remote).

1. Download the netplan configuration file

```
sudo wget -O /etc/netplan/40-cx7.yaml https://github.com/NVIDIA/dgx-spark-
playbooks/raw/main/nvidia/connect-two-sparks/assets/cx7-netplan.yaml
```

2. Set appropriate permissions on the configuration file

```
sudo chmod 600 /etc/netplan/40-cx7.yaml
```

3. Apply the netplan configuration

```
sudo netplan apply
```

2.5.3.1.3.2 Option 2: Manual IP assignment (advanced)

Follow these steps to manually assign IP addresses for dedicated cluster networking.

1. On Node 1, assign a static IP address and bring up the interface

```
sudo ip addr add 192.168.100.10/24 dev enP2p1s0f1np1
sudo ip link set enP2p1s0f1np1 up
```

2. On Node 2, assign a static IP address and bring up the interface

```
sudo ip addr add 192.168.100.11/24 dev enP2p1s0f1np1
sudo ip link set enP2p1s0f1np1 up
```

3. From Node 1, verify connectivity by testing connection to Node 2

```
ping -c 3 192.168.100.11
```

4. From Node 2, verify connectivity by testing connection to Node 1

```
ping -c 3 192.168.100.10
```

2.5.3.1.4 Run the DGX Spark Discovery Script

This step will automatically identify interconnected DGX Spark systems and set up SSH authentication without requiring a password.

The following commands should be run in a terminal session (either local or remote) on both nodes.

On both nodes:

1. Download the discovery script

```
wget https://github.com/NVIDIA/dgx-spark-playbooks/raw/refs/heads/main/
↪nvidia/connect-two-sparks/assets/discover-sparks
```

2. Make the script executable

```
chmod +x discover-sparks
```

3. Run the discovery script

```
./discover-sparks
```

Example output:

```
Found: 192.168.100.10 (spark-1b3b.local)
Found: 192.168.100.11 (spark-1d84.local)
```

Copying your SSH public key to all discovered nodes using ssh-copy-id.

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```
You may be prompted for your password on each node.
Copying SSH key to 192.168.100.10 ...
Copying SSH key to 192.168.100.11 ...
nvidia@192.168.100.11's password:

SSH key copy process complete. These two sparks can now talk to each
→ other.
```

2.5.3.1.5 Install Required Software and Verify the Configuration

With the networking configured and the systems able to communicate with each other, the next step is to install the required software for distributed workloads and run test workloads to verify that GPU-to-GPU communication is working correctly and to measure performance across the stacked systems.

For complete instructions on building NCCL, running the NCCL test suite, and interpreting the results, see the [NCCL Stacked Sparks](#) playbook.

2.5.3.1.6 Troubleshooting

- ▶ Ensure the QSFP/CX7 interface is active and used for IP assignment
- ▶ Verify connectivity between nodes via `ping`
- ▶ Check your interface bindings with `ip a` and `ethtool`
- ▶ If the discovery script fails, manually verify SSH connectivity between nodes
- ▶ For additional troubleshooting guidance and support options, see [spark-maintenance-troubleshooting](#)

2.5.3.1.7 Next Steps

Once tested, this configuration can be scaled to support:

- ▶ Job orchestration with Slurm or Kubernetes
- ▶ Containerized execution with Singularity or Docker

2.6. Software

The DGX Spark comes with a comprehensive software stack optimized for AI development, machine learning, and data science workflows. This section provides detailed information about the included software components and their configuration.

2.6.1. DGX OS

2.6.1.1 Overview

NVIDIA DGX OS is a customized Linux distribution that provides a stable, tested, and supported operating system foundation for running AI, machine learning, and analytics applications on DGX systems. It includes platform-specific optimizations, drivers, and diagnostic tools tailored for NVIDIA hardware.

DGX OS serves as the underlying operating system for your DGX Spark, providing:

- ▶ A robust Linux foundation optimized for AI workloads
- ▶ Pre-configured drivers and system settings for NVIDIA hardware
- ▶ Security updates and system maintenance capabilities
- ▶ Compatibility with the broader NVIDIA software ecosystem

Note

For more information about DGX OS, see the official documentation at: <https://docs.nvidia.com/dgx/dgx-os-7-user-guide/introduction.html>

2.6.1.2 Release Cadence

DGX OS follows a regular release schedule with updates typically provided twice per year, around February and August, for the first two years after initial release. Additional updates and security patches are provided between major releases and throughout the support lifecycle.

2.6.2. NVIDIA Sync

2.6.2.1 Overview

NVIDIA Sync is a system tray utility that provides a simple way to access your DGX Spark system remotely from another machine. Whether your DGX Spark is connected to a monitor or not, NVIDIA Sync makes it easy to connect your favorite development tools and access the system from anywhere on your network.

2.6.2.2 Installation

1. Download the latest version of NVIDIA Sync from <https://build.nvidia.com/spark/connect-to-your-spark/sync>. Installers are available for Windows, macOS, and Linux.
2. Run the installer.

2.6.2.3 Connecting to Your Spark

The first time you launch NVIDIA Sync, you'll be prompted to connect to your DGX Spark.

Provide your DGX Spark's *machine name* and your *login credentials*.

About the machine name

The machine name is your DGX Spark hostname with *.local* appended, such as *spark-xxxx.local*. You can find the default hostname on the Quick Start Guide that came in the box.

The *.local* address uses mDNS (multicast DNS) to automatically locate your DGX Spark on the network without needing to know its IP address. This is particularly useful if your router periodically reassigns IP addresses.

For Windows users: mDNS requires Bonjour Print Services from Apple. If you have iTunes or other Apple software installed, you likely already have this. Otherwise, you can download it from Apple's website. Alternatively, you can try using just the hostname without *.local* (such as *spark-xxxx*), though this method is less reliable on modern networks.

Why .local might not work: *.local* hostnames may not work in enterprise networks with strict security policies, networks that block multicast traffic, or other restricted network environments.

Using an IP address instead: If *.local* hostnames do not work, you will need to use the IP address. To find the IP address, physically log in to your DGX Spark and click the network icon in the top right corner of the Ubuntu desktop. Select **Settings** from the dropdown menu, then navigate to the **Network** section. The IP address will be displayed under your active connection. Alternatively, you can log in to your router's administration console to view connected devices and their IP addresses.

2.6.2.4 Configuring Applications

After connecting to your DGX Spark, NVIDIA Sync will automatically scan your local machine for compatible applications that can connect remotely to the DGX Spark. Select the applications you want to configure.

The selected applications will be configured to use your DGX Spark as a remote compute resource. This means you run the application interface locally on your workstation, but all compute workloads (such as code execution, model training, and data processing) run on the DGX Spark. Your files, code, and development environment are synchronized to the DGX Spark, giving you the full power of the DGX Spark's GPUs while using familiar local tools.

All configured applications are accessible from the NVIDIA Sync system tray icon. You can change your application selections at any time from the NVIDIA Sync settings window. If an application is marked as "unavailable" in the settings, it means the application is not installed on your local machine.

2.6.2.4.1 Available Applications

The following applications can be detected and configured by NVIDIA Sync if they are installed on your local machine:

- ▶ **NVIDIA AI Workbench:** Download from <https://www.nvidia.com/workbench>
- ▶ **Cursor:** Download from <https://cursor.com/downloads>
- ▶ **VS Code:** Download from <https://code.visualstudio.com/download>

The following connection methods are always available and do not require any local application installation:

- ▶ **DGX Dashboard** - Web-based system monitoring and management interface
- ▶ **SSH Terminal** - Command-line access with SSH keys automatically managed by NVIDIA Sync

2.6.3. DGX Dashboard

2.6.3.1 Overview

The DGX Spark comes with a built-in dashboard that provides an overview of the system's current operational metrics, the ability to apply updates, change some system settings, and access local Jupyter Notebooks.

Note

To run updates and change the device name, you must have `sudo` access. The account created during initial setup will have access.

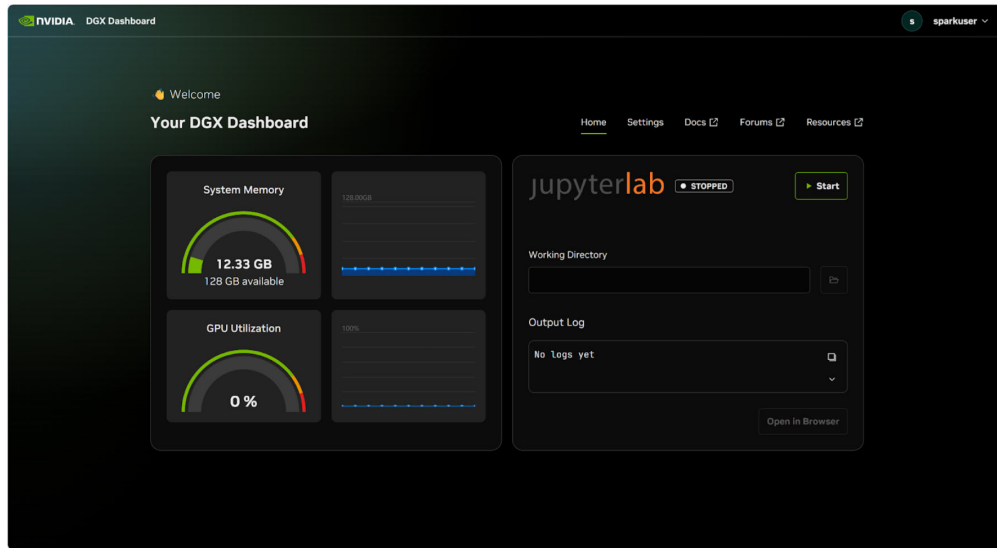


Fig. 2: The DGX Dashboard provides real-time system monitoring and integrated JupyterLab access

2.6.3.2 Integrated JupyterLab

The dashboard includes an integrated JupyterLab instance that provides a convenient development environment:

- ▶ When started, JupyterLab creates a virtual environment in the specified working directory and automatically installs a set of recommended packages
- ▶ If you enter a new working directory and start JupyterLab, a new environment will be created
- ▶ Each user account on the device is assigned a port located in `/opt/nvidia/dgx-dashboard-service/jupyterlab_ports.yaml`
- ▶ To access JupyterLab remotely, you must tunnel it just like the dashboard itself. The port to tunnel is in the ports file. Using NVIDIA Sync, this tunnel is managed for you automatically and just works

2.6.3.3 Accessing the Dashboard

The dashboard can be accessed locally by clicking on the “Show Apps” button in the bottom left corner of the Ubuntu desktop. Then, in the app grid, select the “DGX Dashboard” shortcut to open the dashboard in your default web browser.

Remotely, the dashboard can be accessed using NVIDIA Sync or via a manually created SSH tunnel. If using NVIDIA Sync, after connecting, simply click on the “DGX Dashboard” button and the dashboard will open in your default web browser at <http://localhost:11000>.

To manually access over SSH, first open a tunnel, e.g., `ssh -L 11000:localhost:11000 <username>@<IP or spark-abcd.local>`. Then, open the dashboard in your web browser at <http://localhost:11000>.

2.6.4. NVIDIA Container Runtime for Docker

2.6.4.1 Overview

The NVIDIA Container Runtime enables Docker containers to access GPU resources on DGX Spark systems. It provides hooks based on the Open Container Initiative (OCI) specification, which is an open standard for container runtimes. These OCI hooks are triggered when the `--gpus` flag is used, ensuring GPU devices are made available inside the container. The runtime acts as a bridge between Docker and the NVIDIA drivers, allowing containers to utilize GPU acceleration for AI/ML workloads, CUDA applications, and other GPU-accelerated software.

Key benefits: - Seamless GPU access within containers - Automatic driver and library management - Support for multi-GPU configurations - Compatibility with popular container orchestration platforms

The runtime works in conjunction with the NVIDIA Container Toolkit, which provides the necessary components to expose GPU devices and CUDA libraries to containerized applications.

2.6.4.2 Installation

The NVIDIA Container Toolkit is preinstalled and configured on DGX Spark systems. This includes:

- ▶ NVIDIA Container Runtime
- ▶ Docker integration
- ▶ GPU device access configuration
- ▶ CUDA library management

The runtime is ready to use out of the box for running GPU-accelerated containers.

2.6.4.2.1 Optional: Add User to Docker Group

By default, Docker requires `sudo` privileges to run commands. Adding your user to the `docker` group allows you to run Docker commands without `sudo`, which provides:

- ▶ **Convenience:** No need to type `sudo` before every Docker command
- ▶ **Better workflow:** Seamless integration with development tools and scripts
- ▶ **Reduced friction:** Faster iteration when working with containers

To add your user to the `docker` group:

```
sudo usermod -aG docker $USER
newgrp docker
```

Note

This step is optional. You can continue using Docker with `sudo` if you prefer not to modify group memberships.

2.6.4.3 Usage

2.6.4.3.1 Basic GPU Access

Run a container with GPU access using the `--gpus` flag:

```
docker run -it --gpus=all nvcr.io/nvidia/cuda:13.0.1-devel-ubuntu24.04 nvidia-
→smi
```

This command: - Runs an interactive container (-it) - Enables access to all GPUs (--gpus=all) - Uses the NVIDIA CUDA development image - Executes nvidia-smi to display GPU information

2.6.4.3.2 Set GPU Capabilities

Control which GPU capabilities are available to the container:

```
docker run -it --gpus '"capabilities=compute,utility"' nvcr.io/nvidia/cuda:13.
→0.1-devel-ubuntu24.04 nvidia-smi
```

2.6.4.4 Validation

2.6.4.4.1 Test GPU Access

Run the test command to verify GPU access:

```
docker run -it --gpus=all nvcr.io/nvidia/cuda:13.0.1-devel-ubuntu24.04 nvidia-
→smi
```

Expected output should show:

- ▶ GPU device information
- ▶ Driver version
- ▶ CUDA version
- ▶ Memory usage and temperature

2.6.4.5 Troubleshooting

2.6.4.5.1 Runtime Not Found

If you encounter “runtime not found” errors:

1. Verify NVIDIA Container Toolkit is installed:

```
nvidia-ctl --version
```

2. Check Docker daemon configuration:

```
cat /etc/docker/daemon.json
```

3. Restart Docker service:

```
sudo systemctl restart docker
```

2.6.4.5.2 Driver/Container CUDA Mismatch

If you see CUDA version mismatches:

1. Check host CUDA driver version:

```
nvidia-smi
```

2. Use a container image with compatible CUDA version:

```
docker run -it --gpus=all nvcr.io/nvidia/cuda:13.0.1-devel-ubuntu24.04
→nvidia-smi
```

2.6.4.5.3 Permission Issues

If you encounter permission errors:

1. Ensure your user is in the docker group (if not using sudo):

```
groups $USER
```

2. Check device permissions:

```
ls -la /dev/nvidia*
```

3. Verify Docker daemon has access to GPU devices:

```
sudo docker run -it --gpus=all nvcr.io/nvidia/cuda:13.0.1-devel-ubuntu24.
→04 nvidia-smi
```

2.6.4.5.4 Container Startup Issues

If containers fail to start:

1. Check Docker logs:

```
docker logs <container_id>
```

2. Verify GPU devices are available on host:

```
ls /dev/nvidia*
```

3. Test with a minimal container:

```
docker run --rm --gpus=all nvcr.io/nvidia/cuda:13.0.1-devel-ubuntu24.04
→echo "GPU test successful"
```

2.6.5. NGC

2.6.5.1 Overview

NVIDIA GPU Cloud (NGC) is a comprehensive registry of GPU-optimized containers, pre-trained models, and AI/ML software that enables rapid development and deployment of AI applications. For DGX Spark users, NGC provides access to the latest frameworks, tools, and optimized environments specifically designed for the Grace Blackwell architecture.

Key benefits for DGX Spark users:

- **Optimized Containers:** Pre-configured environments with the latest AI/ML frameworks, CUDA, and libraries optimized for Grace Blackwell GPUs

- ▶ **Pre-trained Models:** Access to state-of-the-art models and model collections for various AI tasks
- ▶ **Rapid Development:** Skip complex environment setup and focus on your AI/ML projects
- ▶ **Cutting-edge Software:** Access to the latest NVIDIA software stack and experimental features

NGC is particularly valuable for DGX Spark users because it provides the most current and optimized software stack for this new platform, ensuring you have access to the latest performance optimizations and features.

2.6.5.2 Getting Started

2.6.5.2.1 Create an NGC Account

1. Visit the [NGC website](#)
2. Click **Sign Up** and create a free account
3. Verify your email address
4. Complete your profile information

2.6.5.2.2 Generate an API Key

1. Log in to your NGC account
2. Navigate to **Setup -> API Key**
3. Click **Generate API Key**
4. Copy and securely store your API key

Note

Your API key is required for pulling containers and accessing NGC resources. Keep it secure and never share it publicly.

2.6.5.2.3 Install NGC CLI (Optional)

The NGC CLI provides convenient command-line access to NGC resources. Your DGX Spark system requires the ARM64 version of the NGC CLI which can be found on the **ARM64 Linux** tab at <https://org.ngc.nvidia.com/setup/installers/cli>.

For more information on installing and using the NGC CLI, see [NGC CLI Documentation](#).

2.6.5.2.4 Authenticate with Docker

Configure Docker to access NGC registries:

```
# Login to NGC with Docker
docker login nvcr.io
# Username: $oauthtoken
# Password: <your-api-key>
```

2.6.5.3 Basic Usage

2.6.5.3.1 Pull and Run a Container

Start with a popular AI/ML framework container:

```
# Pull a PyTorch container optimized for Grace Blackwell
docker pull nvcr.io/nvidia/pytorch:24.08-py3

# Run the container with GPU access
docker run -it --gpus=all nvcr.io/nvidia/pytorch:24.08-py3
```

2.6.5.3.2 Explore Available Resources

Browse NGC resources through the web interface:

- ▶ **Containers:** AI/ML frameworks, development environments, and specialized tools
- ▶ **Models:** Pre-trained models for computer vision, natural language processing, and more
- ▶ **Helm Charts:** Kubernetes deployment configurations
- ▶ **Jupyter Notebooks:** Interactive tutorials and examples

2.6.5.4 Common Workflows

2.6.5.4.1 Development Environment

Use NGC containers as your development environment:

```
# Run a development container with persistent storage
docker run -it --gpus=all \
-v /path/to/your/project:/workspace \
nvcr.io/nvidia/pytorch:24.08-py3
```

2.6.5.4.2 Model Inference and Training

Access pre-trained models and training scripts:

```
# Pull a model from NGC
ngc registry model download-version nvidia/bert-base-uncased:1

# Or use models directly in containers
docker run -it --gpus=all --runtime=nvidia \
nvcr.io/nvidia/pytorch:24.08-py3
```

2.6.5.5 Best Practices

2.6.5.5.1 Container Management

- ▶ **Pin Versions:** Use specific container tags for reproducible environments
- ▶ **Regular Updates:** Periodically update to newer container versions for latest optimizations
- ▶ **Resource Limits:** Set appropriate memory and CPU limits for your workloads

2.6.5.5.2 Data Persistence

- **Volume Mounts:** Mount your data directories into containers for persistence
- **Model Storage:** Store trained models and checkpoints outside containers
- **Configuration:** Keep configuration files in version control

2.6.5.5.3 Security

- **API Key Security:** Store your NGC API key securely and rotate it regularly
- **Container Scanning:** Scan containers for vulnerabilities before use
- **Network Security:** Use appropriate network configurations for your environment

2.6.5.6 Troubleshooting

2.6.5.6.1 Common Issues

Authentication Failures

```
# Verify your API key is correct
docker login nvcr.io
# Check if your account has access to the requested resource
```

Container Pull Issues

```
# Check network connectivity to NGC registry
curl -I https://nvcr.io

# View container catalog on NGC website
# Visit https://catalog.ngc.nvidia.com/containers

# Or use NGC CLI to list available containers
ngc registry image list nvidia/*

# Try pulling with verbose output to see specific error
docker pull nvcr.io/nvidia/pytorch:24.08-py3
```

GPU Access Problems

```
# Verify NVIDIA Container Runtime is installed
docker run --rm --runtime=nvidia --gpus=all nvcr.io/nvidia/cuda:13.0.1-devel-
→ubuntu24.04 nvidia-smi
```

2.6.5.6.2 Getting Help

- **NGC Documentation:** Visit the [NGC documentation](#)
- **Community Forums:** Join the [NVIDIA Developer Forums](#)
- **Additional Support:** For troubleshooting guidance and support options, see [spark-maintenance-troubleshooting](#)

2.6.6. NVIDIA AI Enterprise for DGX Spark Quickstart Guide

NVIDIA AI Enterprise for DGX Spark is an enterprise-grade software platform for AI development, deployment, and optimization on DGX Spark and NVIDIA GB10 Grace Blackwell Superchip-based systems. It provides libraries, frameworks, microservices, drivers, and enterprise support to accelerate inference, training, customization, observability, and guardrailings.

2.6.6.1 Set Up Your DGX Spark System

Before accessing NVIDIA AI Enterprise for DGX Spark, complete the initial setup of your DGX Spark system. Follow the steps in *First Boot and Initial Configuration*. After your system is powered on and connected to your network, continue with the steps in this guide.

2.6.6.2 Order Confirmation Message

After your order is processed, you will receive an order confirmation message from NVIDIA containing the information required to access NVIDIA AI Enterprise software and enterprise support.

Your NVIDIA Entitlement Certificate is attached to the message and has your product activation keys and instructions.

If you manage software access or licensing in your organization, follow the instructions in the NVIDIA Entitlement Certificate. Otherwise, forward the order confirmation message and NVIDIA Entitlement Certificate to your IT administrator.

2.6.6.3 Create Your NVIDIA AI Enterprise Account

To access NVIDIA AI Enterprise for DGX Spark and technical support, you must have an NVIDIA Enterprise Account, which provides access to:

- ▶ **NVIDIA NGC** - Access to NVIDIA AI Enterprise for DGX Spark software components
- ▶ **NVIDIA Enterprise Support Portal** - Access to support services

Both can be accessed from the [NVIDIA Application Hub](#).

Before you begin, have your order confirmation message available.

1. In the NVIDIA Entitlement Certificate instructions, follow the **Register** link.
2. Fill out the form on the NVIDIA Enterprise Account Registration page and click **REGISTER**. You will receive an email with instructions to log in to the [NVIDIA Application Hub](#).
3. Open the email and click **Log In**.
4. On the Login page, enter your email address and click **Sign In**.
5. On the Create Your Account page, provide and confirm a password, then click **Create Account**. You will receive an email to verify your address.
6. Open the verification email and click **Verify Email Address**.

You can now log in to the [NVIDIA Application Hub](#) to access NVIDIA AI Enterprise for DGX Spark and technical support.

2.6.6.4 Access NVIDIA AI Enterprise for DGX Spark Software Assets

1. Go to the [NVIDIA NGC Catalog](#) and log in if prompted.
2. On the **NVIDIA NGC: AI Development Catalog** page, set the **NVIDIA AI Enterprise DGX Spark** filter.
3. Review the filtered software assets.
4. Click a software asset to learn more or download it.

2.7. Common Use Cases

The DGX Spark is designed to support a wide range of AI, machine learning, and data science workflows. Visit <https://build.nvidia.com/spark> for practical guides to help you get started. That site is regularly updated with new content and information and will be the single source of truth for your Spark device.

2.8. PXE Boot Setup

The DGX Spark UEFI BIOS supports PXE boot. Several manual customization steps are required to get PXE to boot the DGX OS image or the DGX Spark recovery image.

Caution

This document is meant to be used as a reference. Explicit instructions are not given to configure the DHCP, HTTP, and TFTP servers. The end user's IT team is expected to configure these servers to fit their company's environment and security guidelines.

2.8.1. Requirements

- ▶ TFTP server
 - ▶ Software that provides TFTP service.
- ▶ HTTP server
 - ▶ An HTTP server is used to transfer large files, such as the iso image and `initrd`. alternatively, TFTP can be used for this purpose. HTTP is used in the example below.
- ▶ DHCP server
 - ▶ Software that provides dynamic host configuration protocol (DHCP) service.

Note

The TFTP server, HTTP server, and DHCP server can all be configured on the same system, or they can each be on different systems.

- ▶ linux bootloader
- ▶ ip address: `<ftp ip>`

- fully qualified host: <ftp host>

This topic provides some guidance concerning how to set up a PXE boot environment for DGX systems. For complete details, refer to online documentation for setting up a PXE boot server. In this example, `xinetd` is used to provide TFTP service; `dnsmasq` is used to provide DHCP service; and `syslinux` is used as the bootloader.

2.8.2. Overview of the PXE Server

The PXE server requires configuration in the following areas:

- bootloader (grub)
- TFTP contents (the kernel and `initrd`)

In this example, TFTP is configured to serve files from `/local/tftp/`. You will need to configure your TFTP server to serve files from `/local/tftp` or the directory you desire to use.
- HTTP contents (the iso image)

In this example, HTTP is configured to serve files from `/local/http/`. You will need to configure your HTTP server to serve files from `/local/http` or the directory you desire to use.
- DHCP

2.8.2.1 PXE Server Configuration for DGX OS Image

In this example, the directory structure on the HTTP and TFTP server looks like this:

```
/local/
  http/
    base_os_7.0.0/
      base_os_7.0.0.iso

  tftp/
    baseos/
      vmlinuz
      initrd
    grub/
      grub.cfg
      grubnetaa64.efi.signed
```

Note

The `vmlinuz` and `initrd` files are specified relative to the TFTP root, `/local/tftp/`; and the location of the ISO, `base_os_7.0.0.iso`, is relative to the HTTP root, `/local/http/`.

Here, the DHCP and PXE servers are configured to use the above directory structure. The person responsible for deploying the PXE environment should change the directory names and structure to fit their infrastructure.

You can set up the directory structure on your HTTP and TFTP server similarly.

The contents of the `/local/tftp/grub/grub.cfg` file should look something like this:

```

set default=0
set timeout=5

menuentry 'Install BaseOS 7.0.0' {
    linux /baseos/vmlinuz fsck.mode=skip autoinstall ip=dhcp url=http://
    ↪<Server IP>/base_os_7.0.0/base_os_7.0.0.iso nvme-core.multipath=n nouveau.
    ↪modeset=0
    initrd /baseos/initrd
}

```

Note

The kernel boot parameters should match the contents of the corresponding ISO's boot menu found in `/mnt/boot/grub/grub.cfg`.

When the system being installed boots via PXE, boot files located on `/local/tftp` are retrieved from the TFTP server. (In this example, the TFTP server is provided by the `xinetd` service whose configuration file, `/etc/xinetd.d/tftp`, specifies that the boot files are located on `/local/tftp`.) When a system is PXE booted, the `grubnetaa64.efi.signed` file that is designated in the DHCP server's `dhcpd.conf` file is retrieved by TFTP transfer (see [Configure Your DHCP Server](#)). By default, after the `grubnetaa64.efi.signed` is booted, the PXE boot `grub.cfg` file, `grub/grub.cfg` in this example, provides menu options for booting further. The PXE boot `grub.cfg` config file specifies the locations of the kernel and `initrd` files relative to the tftp directory.

► Configure the HTTP Directory:

Configure the HTTP file directory and ISO image by placing a copy of the Base OS 7.0.0 ISO in directory `/local/http/base_os_7.0.0/`. In this example, the full path is `/local/http/base_os_7.0.0/base_os_7.0.0.iso`.

► Configure the TFTP Directory By Using the Following Steps:

Mount the Base OS 7.0.0 ISO. Assume your mount point is `/mnt`:

```
sudo mount -o loop /local/http/base_os_7.0.0/base_os_7.0.0.iso /mnt
```

Copy the kernel and `initrd` from the ISO to the tftp directory:

```
cp /mnt/casper/vmlinuz /local/tftp/baseos/vmlinuz
cp /mnt/casper/initrd /local/tftp/baseos/initrd
```

Unmount the Base OS 7.0.0 ISO:

```
umount /mnt
```

► Download GRUB binaries and Copy it for PXE Booting into Place:

```

cd /tmp
wget -q http://ports.ubuntu.com/ubuntu-ports/dists/jammy/main/uefi/grub2-
    ↪arm64/current/grubnetaa64.efi.signed
cp grubnetaa64.efi.signed /local/tftp/

```

2.8.2.2 PXE Server Configuration for Recovery Media

In this example, the directory structure on the HTTP and TFTP server looks like this:

```
/local/
  http/
    fastos/
      usb.customer.tar.gz
  tftp/
    fastos/
      vmlinuz
      initrd
    grub/
      grub.cfg
      grubnetaa64.efi.signed
```

Note

The `vmlinuz` and `initrd` files are specified relative to the TFTP root, `/local/tftp/`; and the location of the ISO, `usb.customer.tar.gz`, is relative to the HTTP root, `/local/http/`.

Here, the DHCP and PXE servers are configured to use the above directory structure. The person responsible for deploying the PXE environment should change the directory names and structure to fit their infrastructure.

You can set up the directory structure on your HTTP and TFTP server similarly.

The contents of the `/local/tftp/grub/grub.cfg` file should look something like this:

```
set default=0
set timeout=5

menuentry "Install DGX Spark FastOS" {
    linux /fastos/vmlinuz nouveau.modeset=0 console=tty0 console=ttyS0,921600
    ↪sbsa_gwdt.action=1 noui pxeinstall=true fastos_usbimg_url=http://<Server IP>
    ↪/fastos/usb.customer.tar.gz ip=dhcp static_ip=<static ip>:<gateway ip>
    initrd /fastos/initrd
}
```

When the system being installed boots via PXE, boot files located on `/local/tftp` are retrieved from the TFTP server. (In this example, the TFTP server is provided by the `xinetd` service whose configuration file, `/etc/xinetd.d/tftp`, specifies that the boot files are located on `/local/tftp`.) When a system is PXE booted, the `grubnetaa64.efi.signed` file that is designated in the DHCP server's `dhcpd.conf` file is retrieved by TFTP transfer (see [Configure Your DHCP Server](#)). By default, after the `grubnetaa64.efi.signed` is booted, the PXE boot `grub.cfg` file, `grub/grub.cfg` in this example, provides menu options for booting further. The PXE boot `grub.cfg` config file specifies the locations of the kernel and `initrd` files relative to the `tftp` directory.

► Configure the HTTP Directory:

Configure the HTTP file directory and ISO image by placing a copy of the DGX OS exactly as `/local/http/usb.customer.tar.gz`. Download the recovery media archive file (tar.gz format) from <https://developer.nvidia.com/downloads/dgx-spark/dgx-spark-recovery-image-1.105.17.tar.gz> and change its name to `usb.customer.tar.gz`.

► Configure the TFTP Directory By Using the Following Steps:

Untar the `usb.customer.tar.gz` file to the `/tmp` directory:

```
tar xpfv /tmp/usb.customer.tar.gz
```

Copy the kernel and `initrd` from the tarball to the `tftp` directory:

```
cp /tmp/usbimg.customer/usb/vmlinuz /local/tftp/fastos/
cp /tmp/usbimg.customer/usb/initrd /local/tftp/fastos/
```

- Download GRUB binaries and Copy it for PXE Booting into Place:

```
cd /tmp
wget -q http://ports.ubuntu.com/ubuntu-ports/dists/jammy/main/uefi/grub2-
→arm64/current/grubnetaa64.efi.signed
cp grubnetaa64.efi.signed /local/tftp/
```

2.8.3. TFTP and HTTP Server Verification

After you have set up all elements of your PXE server, prior to doing a PXE install, verify that the TFTP and HTTP servers are working properly.

TFTP Server Verification

To verify that the TFTP server has been set up correctly, from a different system on the same subnet, use `tftp` to get one of the files that will be obtained via `tftp` during the PXE boot. In this example, the TFTP server has been set up to serve files from `/local/tftp`. The grub configuration file, `grub.cfg` is located on `/local/tftp/grub/grub.cfg`; therefore, from the TFTP command prompt, request `grub.cfg` via `get grub/grub.cfg`.

```
cd /tmp
tftp <TFTP_Server_IP>
get grub/grub.cfg
quit
```

HTTP Server Verification

To verify that the HTTP server has been set up correctly, use the `wget` command to get one of the files that will be obtained via HTTP during the PXE boot. In this example, the HTTP server has been set up to serve files from `/local/http`. The tarball, `usb.customer.tar.gz`, is located on `/local/http/fastos/usb.customer.tar.gz`; therefore, test the HTTP request to retrieve `usb.customer.tar.gz` by running the following commands:

```
cd /tmp
wget http://<HTTP_Server_IP>/fastos/usb.customer.tar.gz
```

2.8.4. Useful Parameters for Configuring Your System's Network Interfaces

`static_ip=<static ip>:<gateway ip>`: tells the `initramfs` to configure the system's interfaces using a static IP address.

2.8.5. Parameters Unique to the Spark OS Installer

- ▶ `usb.skipfw`: Skips the firmware update step during the installation.
- ▶ `usb.shell`: Starts a shell instead of installing the OS.
- ▶ `noui`: Disables the user interface during the installation.
- ▶ `autoinstall`: Enables the autoinstall feature during the installation.

2.8.6. Configure Your DHCP Server

The DHCP server is responsible for providing the IP address of the TFTP server and the name of the bootloader file in addition to the usual role of providing dynamic IP addresses. The address of the TFTP server is specified in the DHCP configuration file as `next-server`, and the bootloader file is specified as `filename`. The architecture option can be used to detect the architecture of the client system and to serve the correct version of the grub bootloader (x86, IA-32, ARM, and so on).

An example of the PXE portion of `dhcpd.conf` is:

```
class "pxeclients" {
    match if substring (option vendor-class-identifier, 0, 9) = "PXEClient";
    filename "grubnetaa64.efi.signed";
    next-server <TFTP_Server_IP>;
    option root-path "/local/tftp/";
}
```

For example, this is a `dhcpd.conf` file:

```
authoritative;
default-lease-time 120;
max-lease-time 120;
ddns-update-style none;

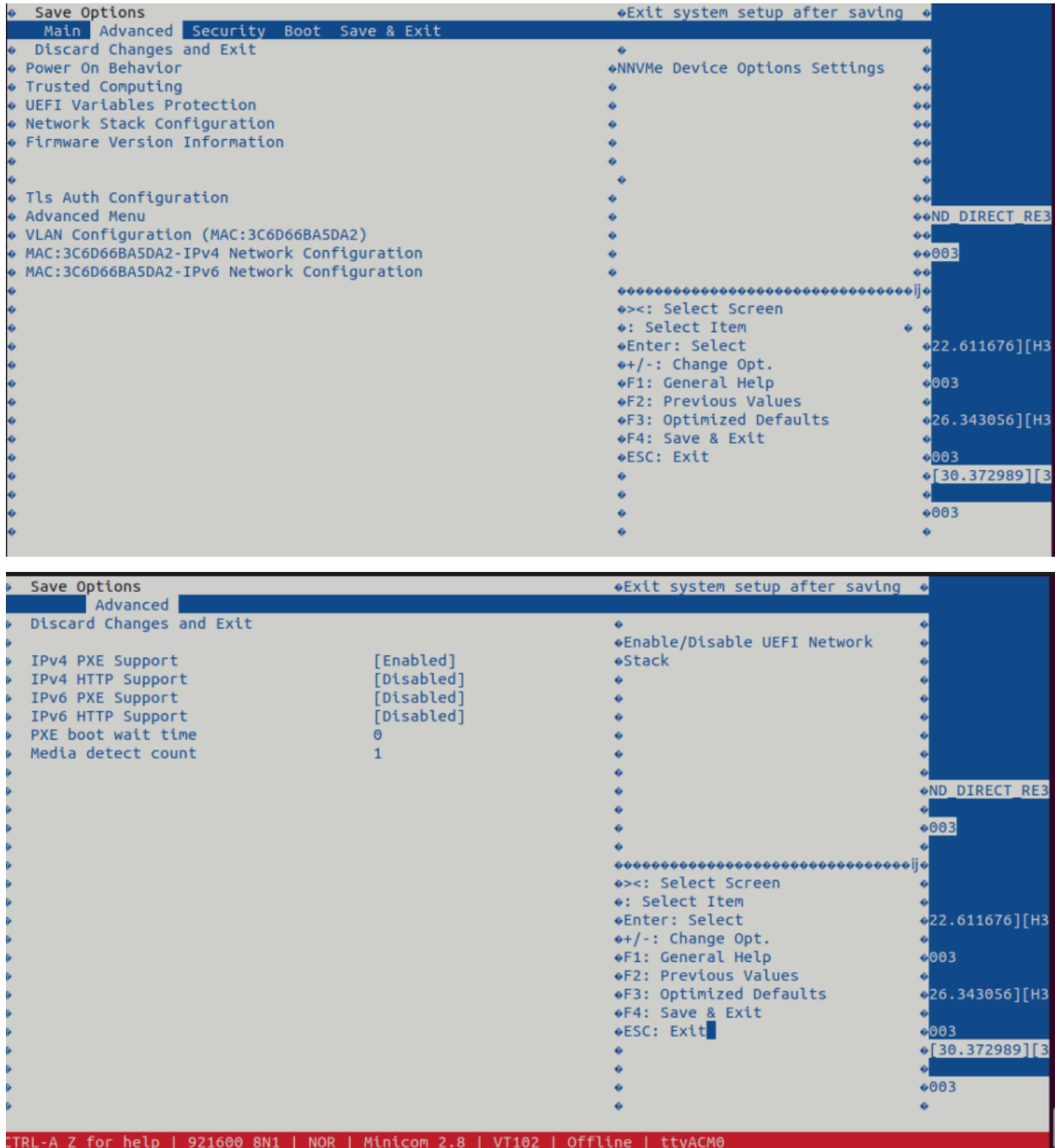
subnet 192.168.99.0 netmask 255.255.255.0 {
    pool {
        range 192.168.99.2;

        # Required for PXE Boot
        class "pxeclients" {
            match if substring (option vendor-class-identifier, 0, 9) = "PXEClient";
            filename "grubnetaa64.efi.signed";
            # TFTP Server IP address
            next-server 192.168.99.1;
            option root-path "/local/tftp/";
        }
    }
}
```

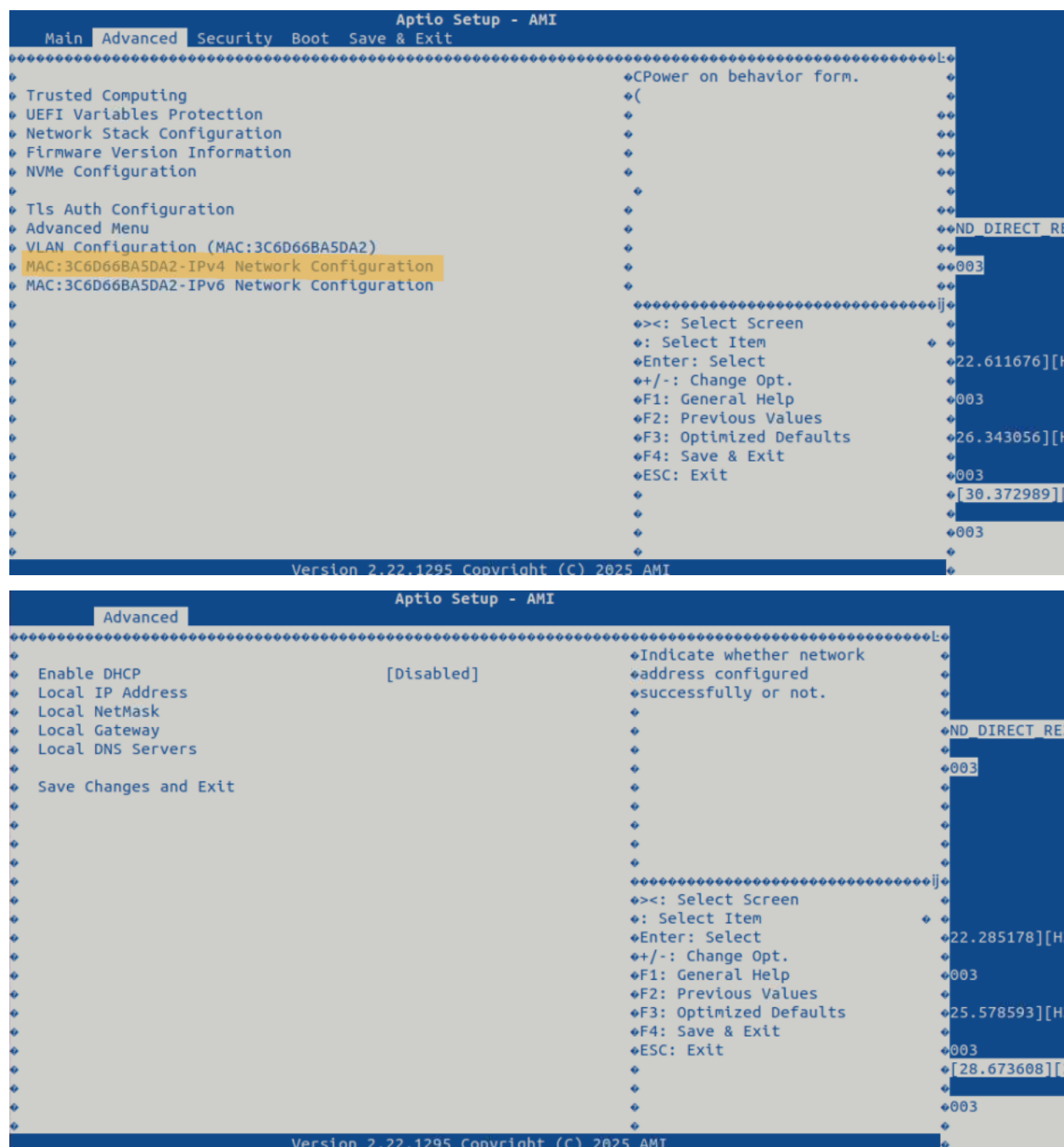
2.8.7. Enable PXE Boot on the DGX Spark

Before enabling PXE Boot, you need to disable the secure boot feature in the UEFI menu or enroll the grubnetaa64.efi.signed in the UEFI menu.

Enable PXE Boot on the DGX Spark by enabling it in Network Stack Configuration menu.



Select the NIC you want to use for PXE Boot and enable PXE Boot.



Save and Exit.

After you finished, you can use the UEFI boot menu to do PXE boot and modify the boot order. See <https://docs.nvidia.com/dgx/dgx-spark-uefi/boot-tab.html>.

2.9. OS and Component Update Guide

This section provides guidance for updating the operating system, software components, and firmware on your DGX Spark. The DGX Spark runs on NVIDIA DGX OS, which is an Ubuntu-based Linux distribution optimized for AI workloads.

Note

Founders Edition Only: The update information in this guide applies only to the DGX Spark Founders Edition. Devices from other manufacturers may have different update procedures.

2.9.1. Update Methods

We strongly recommend using the DGX Dashboard for all system updates on your [spark]. The dashboard ensures your system stays up to date with the latest NVIDIA-optimized components and configurations. While standard Ubuntu package management methods will work, the dashboard provides the most reliable and tested update path for your DGX system.

2.9.2. Using DGX Dashboard for Updates

The DGX Dashboard is the **primary and recommended** way to perform system updates on your DGX Spark. It provides a centralized interface for:

- ▶ Viewing available system updates
- ▶ Installing security patches and system updates
- ▶ Managing NVIDIA driver updates
- ▶ Managing firmware updates
- ▶ Monitoring update status and progress

For detailed information about using the DGX Dashboard, including how to access it and perform updates, see [DGX Dashboard](#).

Warning

Before performing any system or firmware updates:

- ▶ Ensure your system is connected to a stable power source
- ▶ Close all running applications and save your work
- ▶ Have a recovery plan in place
- ▶ Schedule updates during maintenance windows when possible

2.9.3. Manual System Updates

Note: While manual updates using standard Ubuntu package management commands will work, we strongly recommend using the DGX Dashboard for all updates to ensure optimal system performance and compatibility.

For advanced users or when the DGX Dashboard is not available, you can perform system updates manually using the following steps:

1. Open a remote or local terminal on the DGX Spark device.
2. Run the following commands:

```
sudo apt update
sudo apt dist-upgrade
sudo fwupdmgr refresh
sudo fwupdmgr upgrade
sudo reboot
```

These commands will:

- ▶ Update the package lists and upgrade all installed packages (including OS components and drivers)
- ▶ Refresh the firmware metadata and upgrade all firmware components
- ▶ Reboot the system to apply updates

2.9.4. Update Best Practices

- ▶ **Use the DGX Dashboard:** Always prefer the DGX Dashboard for system updates to ensure compatibility and optimal performance
- ▶ **Regular updates:** Check for updates regularly, especially security patches
- ▶ **Backup before major updates:** Always backup critical data before major system changes
- ▶ **Stable power:** Ensure your system has a stable power supply during updates
- ▶ **Maintenance windows:** Schedule updates during planned maintenance windows when possible

2.10. System Recovery

This section provides information about system recovery procedures for your DGX Spark.

Note

Founders Edition Only: This recovery information applies only to the DGX Spark Founders Edition. Devices from other manufacturers may have different recovery procedures.

2.10.1. System Recovery Overview

The system recovery process for the DGX Spark allows you to restore the operating system and firmware to their original factory state. This process is useful when dealing with system corruption, fatal configuration errors, or other issues that prevent normal system operation.

2.10.1.1 Recovery Requirements

Before beginning the recovery process, ensure you have the following:

- ▶ A USB flash drive with 16GB or larger capacity
- ▶ A keyboard and display connected to your DGX Spark
- ▶ Access to download the recovery media from NVIDIA support

2.10.1.2 Recovery Process Steps

The recovery process involves downloading recovery media, creating a bootable USB drive, and following a series of UEFI configuration steps. Follow these steps carefully to ensure successful system recovery.

1. Download Recovery Media:

- ▶ Download the recovery media archive file (tar.gz format) from <https://developer.nvidia.com/downloads/dgx-spark/dgx-spark-recovery-image-1.105.17.tar.gz>
- ▶ Save the file to your local system

2. Create Recovery USB Drive:

- ▶ Download and unzip the recovery media archive file
- ▶ Insert the USB drive into your system
- ▶ Within the extracted files, use one of the following commands to create the recovery drive:

Note

Creating the recovery drive requires administrator privileges. On Windows systems, this is typically done by running the command as an administrator (i.e. from an elevated PowerShell or Command Prompt). If supported by the host system, you will be prompted by the system to run the command with elevated privileges.

Windows:

```
CreateUSBKey.cmd
```

Linux:

```
CreateUSBKey.sh
```

MacOS:

```
CreateUSBKeyMacOS.sh
```

Warning

This process will erase all data on your USB drive. Ensure you have backed up any important data before proceeding.

3. Boot from Recovery USB Drive:

- ▶ Disconnect any external storage devices from the DGX Spark
- ▶ Connect the USB drive to one of the USB ports on your DGX Spark
- ▶ If your DGX Spark is powered off, boot the device into UEFI settings by holding down the Esc or Del key immediately after powering on the device

Important

Make sure to use a keyboard plugged directly into a USB port on the DGX Spark device. Some Bluetooth keyboards, even those with USB connections, may not work during this process. If your keyboard isn't recognized, use a standard USB keyboard.

4. Restore UEFI Defaults:

- ▶ Tap the Right Arrow key to select the "Save & Exit" page within the UEFI settings
- ▶ Tap Down Arrow to select "Restore Defaults" and select "Yes" in response to "Load Optimized Defaults"
- ▶ Select "Save Changes and Reset" - the device will reboot
- ▶ While the device is rebooting, hold down the Esc or Del key to re-enter UEFI settings a second time

5. Enable Secure Boot:

- ▶ Use the Right Arrow key to select "Security"
- ▶ Confirm that "Secure Boot" is set to "Enabled"
- ▶ Select "Restore Factory Keys"
- ▶ Use the arrow keys to select the "Save and Exit" BIOS screen, and select "Save Changes and Reset"

6. Boot from Recovery Media:

- ▶ While the device is rebooting, hold down the Esc or Del key to re-enter UEFI settings a third time
- ▶ Tap the Right Arrow key to select the "Save & Exit" page in UEFI settings
- ▶ Tap Down Arrow to move to the "Boot Override" section
- ▶ Select the USB drive and tap Enter
- ▶ The device will reboot using the USB drive - follow the on-screen steps to update your firmware and re-install the OS on the device's SSD

7. Restore System from Recovery Environment

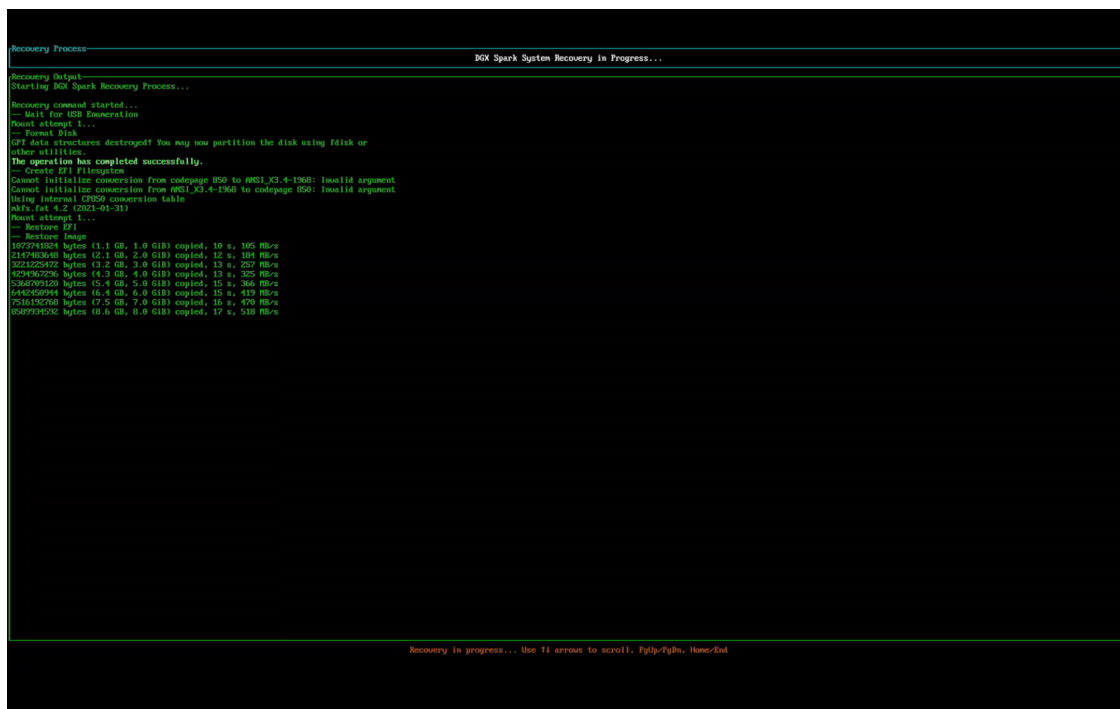
- ▶ On the welcome screen, press Enter to continue. To cancel the process, press Esc.



- On the warning screen, select [EXIT] to restart without proceeding, or select [START RECOVERY] to begin reflashing the SSD. Be aware that this will completely erase the internal SSD on the DGX Spark.



- Monitor the Recovery Progress screen for detailed output from the recovery process and the progress of the SSD reflash.



- ▶ When the process is complete, review the progress screen and follow the prompt to continue. Scroll or page through the recovery output if needed, for example during customer support.



- On the final screen, confirm that the DGX Spark has been reset to factory state and press Enter to restart the device.



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d) Convey the object code by offering access from a designated place (gratis or for a charge), and offer equivalent access to the Corresponding Source in the same way through the same place at no further charge. You need not require recipients to copy the Corresponding Source along with the object code. If the place to copy the object code is a network server, the Corresponding Source may be on a different server (operated by you or a third party) that supports equivalent copying facilities, provided you maintain clear directions next to the object code saying where to find the Corresponding Source. Regardless of what server hosts the Corresponding Source, you remain obligated to ensure that it is available for as long as needed to satisfy these requirements.

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